

3^k FD with three factors

Kontadakis Eftychis

Introduction

In this work we are going to apply a 3^k FD in a 3 factor experiment.

Data

We have 2 replicates in each combination level.

```
# Factors
A <- factor(rep(c("A1","A2","A3"), each = 18))
B <- factor(rep(rep(c("B1","B2","B3"), each = 6), times = 3))
C <- factor(rep(rep(c("C1","C2","C3"), each = 2), times = 9))

Response <- c(
  -35, -25, -45, -60, -40, 15, 17, 24, -65, 58, 20, 4, -39, -35, -55, -67, 15, -30,
  110, 75, -10, 30, 80, 54, 55, 120, -55, -44, 110, 44, 90, 113, -28, -26, 110, 135,
  4, 5, -40, -30, 31, 36, -23, -5, -64, -62, -20, -31, -30, -55, -61, -52, 54, 4)

# Create the dataframe
data <- data.frame(A, B, C, Response)
```

With the dataframe ready we can proceed to running the actual ANOVA and Regression

ANOVA

```
anova_model <- aov(Response ~ A * B * C, data = data)
summary(anova_model)
```

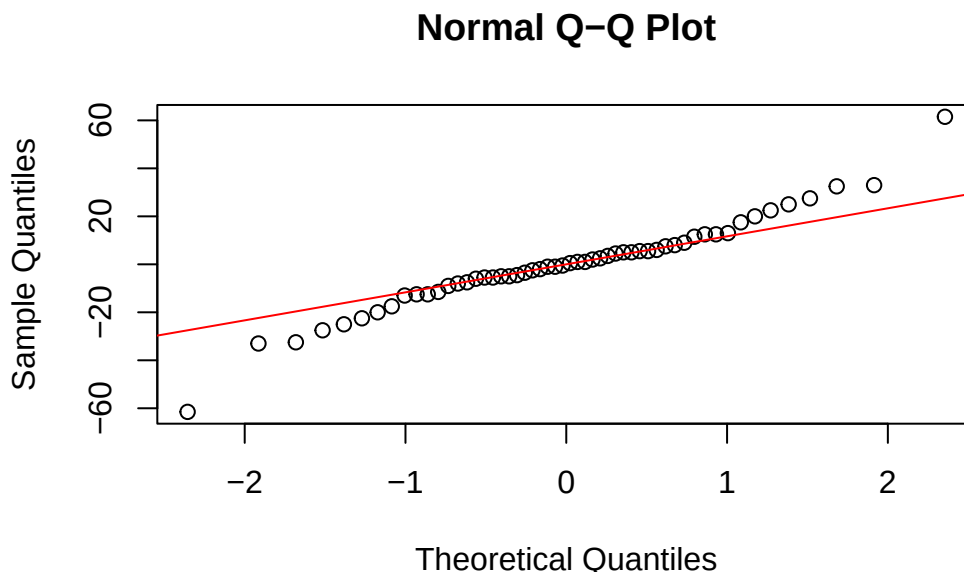
	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
A	2	62979	31489	44.618	2.76e-09	***
B	2	358	179	0.254	0.77785	
C	2	50772	25386	35.970	2.43e-08	***
A:B	4	13318	3330	4.718	0.00512	**
A:C	4	17151	4288	6.075	0.00128	**
B:C	4	5184	1296	1.836	0.15096	
A:B:C	8	4328	541	0.767	0.63461	
Residuals	27	19055	706			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

We can see that the effects A, C and AB and AC are important.

We are of course going to show the Q-Q plot to make sure our original assumptions were correct:

```
# QQ plot
qqnorm(residuals(anova_model))
qqline(residuals(anova_model), col = "red")
```



Regression

```
# Set the orthogonal contrasts:
contrasts(data$A) <- contr.poly(3)
contrasts(data$B) <- contr.poly(3)
contrasts(data$C) <- contr.poly(3)

reg_model <- lm(Response ~ A * B * C, data = data)
summary(reg_model)
```

Call:

```
lm(formula = Response ~ A * B * C, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-61.500	-7.875	0.000	7.875	61.500

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.2037	3.6152	1.439	0.161531
A.L	0.1571	6.2617	0.025	0.980164
A.Q	-59.1506	6.2617	-9.446	4.75e-10 ***
B.L	-4.3998	6.2617	-0.703	0.488289
B.Q	0.7258	6.2617	0.116	0.908584
C.L	8.8388	6.2617	1.412	0.169497
C.Q	52.3692	6.2617	8.363	5.66e-09 ***
A.L:B.L	-10.4167	10.8456	-0.960	0.345350
A.Q:B.L	-13.3272	10.8456	-1.229	0.229745
A.L:B.Q	38.1532	10.8456	3.518	0.001560 **
A.Q:B.Q	-21.8611	10.8456	-2.016	0.053887 .
A.L:C.L	8.4167	10.8456	0.776	0.444467
A.Q:C.L	15.1554	10.8456	1.397	0.173676
A.L:C.Q	11.0178	10.8456	1.016	0.318706
A.Q:C.Q	-49.3611	10.8456	-4.551	0.000102 ***
B.L:C.L	16.8333	10.8456	1.552	0.132284
B.Q:C.L	19.6299	10.8456	1.810	0.081449 .
B.L:C.Q	13.9526	10.8456	1.286	0.209197
B.Q:C.Q	-0.7778	10.8456	-0.072	0.943358
A.L:B.L:C.L	10.7834	18.7851	0.574	0.570693
A.Q:B.L:C.L	-7.8588	18.7851	-0.418	0.678999

A.L:B.Q:C.L	12.1454	18.7851	0.647	0.523386
A.Q:B.Q:C.L	18.2080	18.7851	0.969	0.341009
A.L:B.L:C.Q	-4.7969	18.7851	-0.255	0.800382
A.Q:B.L:C.Q	-31.8787	18.7851	-1.697	0.101193
A.L:B.Q:C.Q	-3.1231	18.7851	-0.166	0.869198
A.Q:B.Q:C.Q	21.3990	18.7851	1.139	0.264647

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

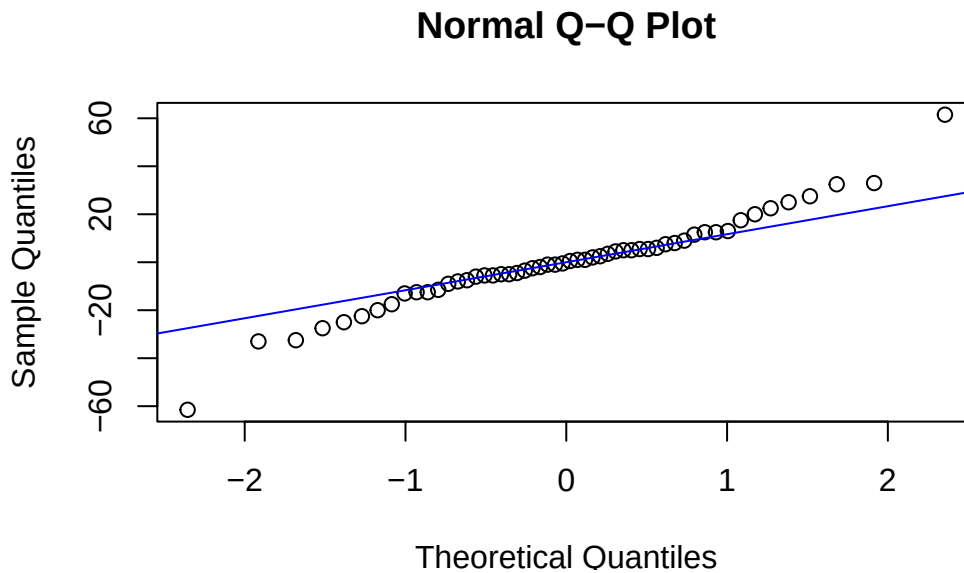
Residual standard error: 26.57 on 27 degrees of freedom

Multiple R-squared: 0.8899, Adjusted R-squared: 0.784

F-statistic: 8.397 on 26 and 27 DF, p-value: 2.251e-07

Again the Q-Q Plot:

```
# QQ plot
qqnorm(residuals(reg_model))
qqline(residuals(reg_model), col = "blue")
```



As expected the Q-Q plots much completely for both the ANOVA and the Regression.